

Protein methionine content and MDA-lysine adducts are inversely related to maximum life span in the heart of mammals.



Aging affects all organisms and its basic mechanisms are expected to be conserved across species. Oxidation of proteins has been proposed to be one of the basic mechanisms linking oxygen radicals with the basic aging process. If oxidative damage to proteins is involved in aging, long-lived animals (which age slowly) should show lower levels of markers of this kind of damage than short-lived ones. However, this possibility has not been investigated yet. In this study, steady-state levels of markers of different kinds of protein damage--oxidation (glutamic and aminoadipic semialdehydes), mixed glyco- and lipoxidation (carboxymethyl- and carboxyethyllysine), lipoxidation (malondialdehydelysine) and amino acid composition--were measured in the heart of eight mammalian species ranging in maximum life span (MLSP) from 3.5 to 46 years. Oxidation markers were directly correlated with MLSP across species. Mixed glyco- and lipoxidation markers did not correlate with MLSP. However, the lipoxidation marker malondialdehydelysine was inversely correlated with MLSP ($r^2=0.85$; $P<0.001$). The amino acid compositional analysis revealed that methionine is the only amino acid strongly correlated MLSP and that such correlation is negative ($r^2=0.93$; $P<0.001$). This trait may contribute to lower steady-state levels of oxidized methionine residues in cellular proteins. These results reinforce the notion that high longevity in homeothermic vertebrates is achieved in part by constitutively decreasing the sensitivity of both tissue proteins and lipids to oxidative damage. This is obtained by modifying the constituent structural components of proteins and lipids, selecting those less sensitive to oxidative modifications.